Original Communication

Vitamins – Wrong Approaches

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Abstract: Deficiencies of essential nutrients have been responsible for many epidemic outbreaks of deficiency diseases in the past. Large observational studies point at possible links between nutrition and chronic diseases. Low intake of antioxidant vitamins e. g. have been correlated to increased risk of cardiovascular diseases or cancer. The main results of these studies are indications that an intake below the recommendation could be one of the risk factors for chronic diseases. There was hardly any evidence that amounts above the RDA could be of additional benefit. Since observational studies cannot prove causality, the scientific community has been asking for placebo-controlled, randomized intervention trials (RCTs). Thus, the consequences of the epidemiological studies would have been to select volunteers whose baseline vitamin levels were below the recommended values. The hypothesis of the trial should be that correcting this risk factor up to RDA levels lowers the risk of a disease like CVD by 20–30 %. However, none of the RCTs of western countries was designed to correct a chronic marginal deficiency, but they rather tested whether an additional supplement on top of the recommended values would be beneficial in reducing a disease risk or its prognosis. It was, therefore, not surprising that the results were disappointing. As a matter of fact, the results confirmed the findings of the observational studies: chronic diseases are the product of several risk factors, among them most probably a chronic vitamin deficiency. Vitamin supplements could only correct the part of the overall risk that is due to the insufficient vitamin intake.

Key words: vitamins, observational studies, randomized clinical trials, chronic diseases, Recommended Dietary Allowance, disease risk

During the late 19th century biochemical research was mainly focused at the unraveling of metabolic pathways. It was believed that carbohydrates, lipids, proteins and salts were the only required components of the diet. In 1881, N. Lunin at the university of Basel published the following feeding experiment with mice:

Two mice fed 2½ months with milk could be set in liberty in very good condition. 4 mice fed with distilled water only died after 3 or 4 days. 5 mice on an ash-free diet survived 11–21 days. 6 mice on the same diet receiving salts lived 20–31 days. He concluded that milk contains besides casein, fat, lactose and salts other substances that are essential for the nutrition. He suggested to trace these substances and to explore their importance for nutrition. However, his findings have been ignored since it was believed that toxins or germs only caused diseases. Even after the discovery and description of vitamins by Casimir Funk in 1912 Emil Abderhalden wrote in 1913: There is no evidence for the hypothesis that there are unknown substances that are essential for survival. Thus, these authors obviously used the wrong approach for identifying essential nutrients.

During the first half of the 20th century vitamin deficiencies could then be linked to a specific disease (Table I).

Since vitamins are part of metabolic pathways, their requirement should be known in order to provide the best possible advice for a healthy diet. Therefore, the US Institute of Medicine started in 1941 to publish dietary reference intakes for vitamins and minerals (DRI) for advising the nation and improving health. These recommendations have been revised roughly every ten years, the last time in 2000. The aim of the

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Table I: Vitamin deficiency diseases.

Vitamin	Disease
Vitamin A	Night blindness
Vitamin D	Rickets
Vitamin C	Scurvy
Vitamin B_1	Beriberi
Vitamin B_{12}	Pernicious anemia
Niacin	Pellagra
Folic acid	Megaloblastic anemia

The relationship between the nutrient and the disease is 1:1; e.g. the deficiency of one vitamin causes one disease.

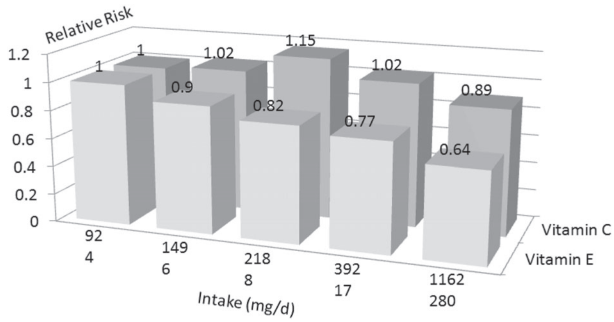

Figure 1: Health Professionals Study: relative risk of CHD in men.

Table II: US RDAs for adult males.

Year	Vitamin A µg	Vitamin E mg	Vitamin C mg	Folic acid µg
1941	1000		75	
1948	1000		75	
1957	1000		70	
1968	1000	30	60	400
1976	1000	15	45	400
1980	1000	10	60	400
1989	1000	10	60	200
2000	700	15	90	400

recommendation was until 1989 the elimination of vitamin deficiency diseases, but the scientific data did not allow to clearly define the requirements. This resulted sometimes in erroneous values as can be seen from Table II.

In the second half of the 20th century, large observational studies consistently pointed at possible links between nutrition and chronic diseases such as low intake of antioxidant vitamins and increased risk of cardiovascular diseases or cancer.

Thus, the scientific rationale for conducting RCTs was excellent and the first outcomes very promising. In the Linxian trial, small but significant reductions of the total and cancer mortality have been observed in participants receiving daily 15 mg β-carotene, 50 µg selenium and 30 mg vitamin E for 5.25 years. The enthusiasm for using antioxidant vitamins as a cheap, nutritional concept for the prevention of chronic diseases weakened when the results of very large randomized intervention trials of western countries became available. The Heart Protection Study e.g. showed neither any benefit nor harm supplementing volunteers a high dose of a combination of vitamin E, vitamin C and β-carotene.

In order to understand the apparent discrepancy between the RCT and the observational studies the major outcomes of the latter will be shortly summarized.

In the WHO – MONICA study the difference of the CVD-mortality between 16 European countries could be explained by 90 % using plasma levels of vitamin E, vitamin C, carotene, cholesterol and blood pressure as predictors. The most important contributor was the plasma level of vitamin E with a range of 19–28 µmol/L between Northern- and Mediterranean countries. Several case control and prospective studies confirmed this relationship; however the risk predic-

Table III: Lessons from epidemiological studies (* = ref 12).

	Favourable plasma level*	Intake (mg/d)*	US-RDA 2000 (mg/d)	DACH 2000 (mg/d)	France 2001 (mg/d)
Vitamin C	50 µmol/L	75–150	♂ : 90; ♀ : 75	100	110
Vitamin E	30 µmol/L	15–30	15 mg/d	♂ : 15–12; ♀ : 12–11	12
β-Carotene	0.4 µmol/L	2–4	♂ : 1.8 ♀ : 1.4 synt.; pro A	2–4	2.1

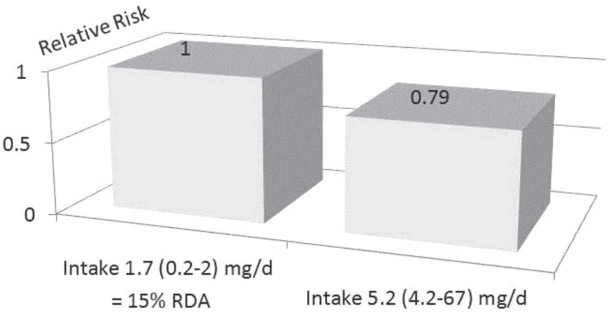

Figure 2: Nurses' Health Study: relative risk of CHD associated with intake of vitamin E.

tion was not as strong. In Scotland, e. g., the risk of angina pectoris was associated with low plasma levels of vitamin E (18.9 μmol/L vs. 28.2 μmol/L), vitamin C (13.1 μmol/L vs. 57.4 μmol/L) and β-carotene (0.26 μmol/L vs. 0.68 μmol/L). Several very large observational studies reported intake data rather than plasma levels. In 39,910 U.S. male health professionals, the multivariate relative risk for coronary disease was 0.64 (95 percent confidence interval [95 % CI], 0.49 to 0.83) for men consuming more than 40 mg per day of vitamin E as compared with those consuming less than 5 mg per day. Vitamin C was not associated with CHD risk, however the consumption in the lowest quintile was 92 mg/d what corresponds to the RDA (Fig 1).

From the figure it is evident that the risk is highest at an intake of 4 mg vitamin E per day that is roughly 25 % of the RDA. The 4th quintile corresponds to one RDA; unfortunately there are no indications whether there is a difference between the 4th and the 5th quintile. The multi-variance analysis reveals the following relative risks (95 % CI):

1st quintile: 1.0; 2nd quintile: 0.90 (0.71 – 1.14); 3rd quintile: 0.82 (0.64 – 1.07); 4th quintile 0.77 (0.60 – 0.98); 5th quintile: 0.64 (0.49 – 0.83); p for trend: 0.003.

A similar study analyzing the dietary behavior of 87,245 female nurses found a relative risk of major coronary disease of 0.79 (95 % CI, 0.61 to 0.1.03) comparing the 5th to the 1st quintile. Although the trend is not significant ($p = 0.12$) probably due to an intake of less than half of the RDA in the 5th quintile, there is still an increased risk at an intake significantly below the RDA (Fig. 2).

Several meta-analyses have combined results from case-control and cohort studies on vitamin E, C and beta.carotene. The outcome can be summarized as follows: studies with a vitamin supply significantly below the RDA are associated with an increased disease risk. If the lowest quintile is close to the RDA, there is no effect. A selection of studies is listed in Fig. 3.

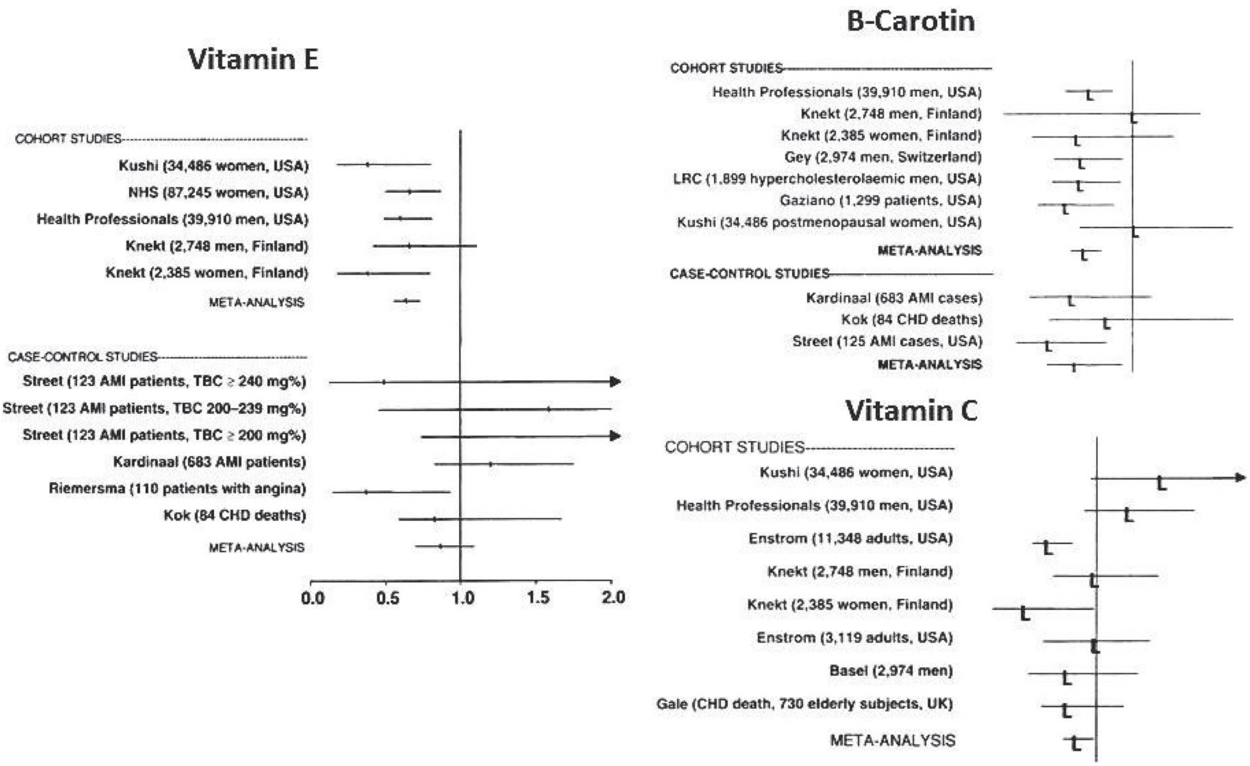

Figure 3: Epidemiolocical Studies, CHD.

Table IV: Plasma concentrations of antioxidants in placebo groups of several randomized trials.

	β-Carotene µmol/L	% of 0.4 µmol/L [a)]	Vitamin E µmol/L	% of 30 µmol/L [a)]	Vitamin C µmol/L	% of 50 µmol/L [a)]
AREDS	0.48	120.0	39.0	130.0	58.1	116.2
ASAP male smoker	0.28	70.0	29.7	99.0	57.4	114.8
ASAP male non-smoker	0.39	97.5	33.5	111.7	68.1	136.2
ASAP female smoker	0.44	110	31.2	104.0	69.8	139.6
ASAP female non-smoker	0.59	147.5	35.4	118.0	82.5	165.0
ATBC	0.34	85.0	28.8	96.0		
CARET	0.32	80.0				
CHAOS			33.2	110.7		
HPS	0.32	80.0	27.0	90.0	43.2	86.4
Linxian: Dysplasia	0.21	52.5			22.5	45.0
Linxian: General	0.22	55.0			30.7	61.4
Skin cancer study	0.39	97.5				
SUVIMAX men	0.56	140.0	31.7	105.7	52.8	105.6
SUVIMAX women	0.76	190.0	31.7	105.7	61.3	122.6

[a)] % of the recommended plasma levels according to ref. 12.

Table V: Major inclusion criteria of some RCTs.

HPS	Adults with coronary disease, other occlusive arterial disease, or diabetes were randomly allocated to receive antioxidant vitamin supplementation or placebo
ASAP	Smoking and nonsmoking men and postmenopausal women with serum cholesterol ≥5.0 mmol/L
HOPE	Women and men who were at high risk for cardiovascular events because they had cardiovascular disease or diabetes in addition to one other risk factor
SUVIMAX	Healthy French adults

In a consensus conference in Hohenheim, researchers came to the conclusion that an adequate plasma level should be achieved in order to benefit from the preventive potential of antioxidant vitamins. The following values have been proposed: 30 µmol/L vitamin E, 50 µmol/L vitamin C and 0.4 µmol/L β-carotene. In order to reach these levels, they suggested an intake of 75–150 mg of vitamin C, 15–30 mg of vitamin E, and 2–4 mg β-carotene per day. These recommendations are in very good agreement with the RDAs in Europe and USA (Table III).

With this lesson in mind, it becomes evident that already the placebo groups of most randomized trials, where plasma levels have been reported, fulfilled this requirement with the exception of the Linxian trial (Table IV).

Thus, observational epidemiological studies always used a reference group that was close to deficiency. Randomized trials, however, answered only the question whether high dose supplements had an additional effect on top of an adequate supply in particular risk groups (Fig 4).

The observational studies point at two major interrelations:

- A vitamin intake significantly below RDA might be a risk factor for chronic diseases
- Chronic diseases are the consequence of the occurrence of several risk factors, among them chronic vitamin deficiency.

To prove this interrelation a RCT should, therefore, test the hypothesis whether the elimination of a chronic vitamin deficiency reduces the risk of a chronic disease. However, from the inclusion criteria of some of the RCTs it is evident that such a hypothesis has, so far, not been tested (Table V). Hence, the wrong approach has been used for the design of RCTs that should confirm findings from observational studies.

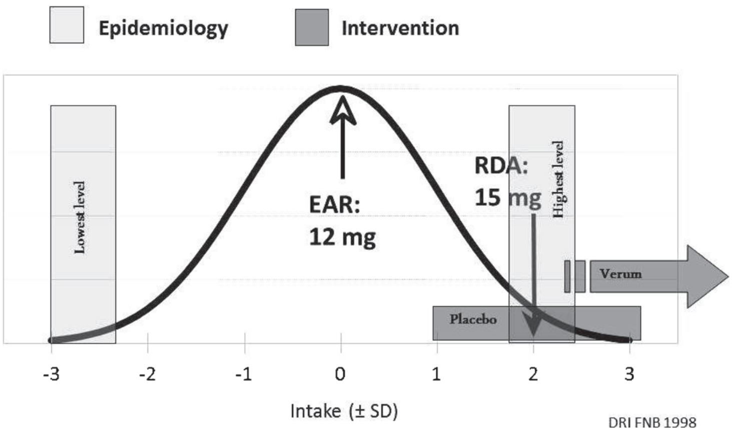

Figure 4: Discrepancy between observational studies and RCTs, e. g. vitamin E.

Conclusion

Wrong approaches have delayed the discovery of vitamins since a group of scientists argued that only germs and toxins can cause diseases, not deficiency of nutrients. Several diseases (scurvy, pellagra, etc.) are now known that are caused by a severe vitamin deficiency. Observational studies point at an association between low intakes of nutrients below RDA and part of the risk for a chronic disease. Although the aim of many RCT was to prove this association, it was only tested whether a high intake of a nutrient on top of an adequate supply (RDA) can reduce the disease risk. Obviously this is not the case, but the question is still open: are chronic vitamin deficiencies a risk factor for chronic diseases? The answer could only be given by correcting low vitamin intakes up to the RDA level and thus lowering the part of the disease risk that is due to the vitamin deficiency. However, this challenge might be too ambitious, more data could be gathered by observational- and mechanistic studies instead.

Abbreviations

AREDS	Age-Related Eye Disease Study
ASAP	Antioxidant Supplementation in Atherosclerosis Prevention study
ATBC	α-Tocopherol, β-Carotene Cancer Prevention Study
CARET	β-Carotene and Retinol Efficacy Trial
CHAOS	Cambridge Heart Antioxidant Study
HOPE	Heart Outcomes Prevention Evaluation Study
HPS	Heart Protection Study
MONICA	Monitoring of Cardiovascular Disease and Cancer
RDA	Recommended Dietary Allowance
SUVIMAX	SUpplémentation en VItamines et Minéraux AntioXydants

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